University of Applied Sciences and Arts Northwestern Switzerland
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# Computacenter

# Repair Process Optimisation

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## 2. Abstract

This teaching case examines the hardware repair process at the Swiss branch of Computacenter to address operational inefficiencies and profitability challenges in the repair services. The study aims to identify pain points and propose optimisation strategies using Business Process Management (BPM) principles.

The methodology includes qualitative research, using interviews and questionnaires to gather insights from stakeholders and employees about the repair process. The information gathered was then used to find solutions and suggestions to help Computacenter to improve its process and to increase BPM maturity.

The results highlight inefficiencies originating from high administrative workloads, reliance on manual tasks, and limited use of automation. Suggestions for improvement include enhanced digitization, integration with external systems, and a cultural shift toward continuous process improvement, which is also known as Kaizen.

The paper concludes that adopting targeted automation and embracing a culture of innovation can significantly improve the repair process. These changes can enhance operational efficiency, reduce costs, and align the repair center with Computacenter's broader strategic goals, providing a scalable and customer-centric approach to device repairs. Furthermore, these solutions will help improve the BPM Maturity in the department.

## 3. Introduction

**Commented [MS1]:** Add the focus on the repair process department

On 12 September 2024, Gil, the Head of Quality Management and Operational Excellence at Computacenter Switzerland, faced a big problem. During talks about a contract with one of the company's longstanding clients, it became clear that the repair process for devices that were still under warranty was not making enough money. More paperwork, the same repair prices and higher Swiss labour costs had together created a big problem for the company. If they don't do something, the company might lose important contracts, which could mean that this part of the business won't be able to keep going.

This problem is from the repair process, which is not efficient. A lot of the work is done by hand, like entering data, dealing with paperwork, and sending messages between different systems. This makes things slower and more expensive and means that the repair centre can't be as competitive as it should be. Gil saw that they needed to use more digital tools and automation to make the processes better, so that they could do less paperwork and be more profitable. But there weren't many resources available to make changes to the company's existing computer system, so it was important to focus on the tasks that would have the biggest impact

## 3.1. Company Profile

Computacenter is an IT Service Provider based in the United Kingdom with approximately 20'000 employees worldwide. It has three major delivery centres Managed Services, Professional Services, and Technology Sourcing. About 10 years ago, Computacenter has acquired a Swiss IT Service provider with approximately 250 employees, who also offers services in the same delivery centres. The local service portfolio however mostly centres around Workplace Lifecycle, whereas the group portfolio also encompasses to cloud, networking, security, application and data centre. Although some supporting processes like local IT and HR have since been aligned or partially integrated to the group, the Swiss business is largely operating independently and uses different processes and tooling to do so.

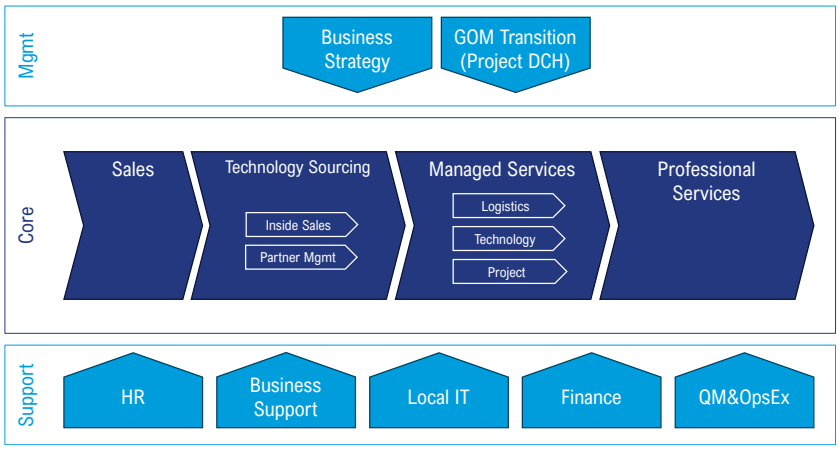


**Figure 1 - Computacenter's Process Landscape**

## 3.2. Scope

This case is specifically focusing on the hardware repair process in Switzerland. This is being performed by the repair centre team. This team is part of workplace maintenance in the Managed Service Delivery centre, as depicted in Figure 1. The repair services of the group are currently out of scope, because they are too dissimilar in structure, scope and tooling. An integration of the Swiss services into the group are not planed in the coming years.

## 3.3. Challenge

Device repairing process in Switzerland branch. This process represents the workflow involving multiple roles: Log inbound, Technicians, Parts Management, Administration, Log outbound.

The current repair process struggles to balance the operational complexity with cost efficiency. The administrative workload per case is very high. Many tasks are carried out manually and involve printing, filling and scanning a lot of paperwork, copy and pasting info between programs and exporting data from the ERP System to excel for processing and reporting. Such

manual tasks require extensive staff leads to high labour cost which impact a profit. The goal of the process is to ensure efficient handling the device repair from receiving the device to sending the device back to the customer, all while maintaining increasing customer satisfaction on functionality of device and reducing the cost and increasing profit.

## 3.4. Method

**Information Collection Approach**

For this case we require and understanding for the process and what steps are involved, the dependencies of each task and the grievances, that the employees within the process have. Having a clear understanding of the process and the wishes of the stakeholders and employees will give us an understanding on how Computacenter can be helped and what improvements can be suggested.

In research there are two overarching research methods: Qualitative and Quantitative. The two approaches differ in what the focus lies and what is the desired outcome. Quantitative research focuses on numeric values and objectives, whereas qualitative research focuses on subjective phenomena. That means with the quantitative research method we would most likely be timing the individual steps and evaluate nominal figures. The qualitative method would lean into conducting interview or surveys to gauge what the stakeholders feel.

We decided upon these two methods to focus on the qualitative research methods. For one we think the people who know the process best are the employees that work the process. These people generally have a good understanding on what should be improved. Additionally, this makes it easier to involve people and help to drive any potential changes forward. On the other hand, we would not be able to gather enough representative data to conduct quantitative research.

To achieve this, we decided on splitting the Data collection into two major steps. First step involves in understanding the process which will require documentation of the process. The process then will be presented to the Head of the department for confirmation and to gather further details.

The second step involves in gathering information from the employees involved in the process. For this we decided on sending two questionnaires. The first is to gather pain points and improvements that employees may have already thought of. The second questionnaire builds upon the first and for one check that our conclusions drawn from the first questionnaire are accurate, and tests on how plausible some improvements may be.

More details and information on the contents of the interview on the questionnaire can be found within the appendix at the end of the document.

**Process Documentation**

We sat down with the department and made an initial documentation of the process on paper. The documentation was then modelled using the BPMN 2.0 Notation. For this the program Camunda was used.

The Process was then discussed within the team so we could gather questions that we would ask the Head of the Department with the goal to clarify the crucial and time-consuming steps

and to get a good understanding what the stakeholders need. These were discussed then in a second interview and used to clarify the process.

**Questionnaire**
With the questionnaire we wanted to gather the knowledge, pain points and wisdom from each role within the Repair Center Process. For this a survey was created within Microsoft forms. For each role there is an introductory question to ask whether the person works in the mentioned role, or whether they have input on these tasks. If they do, the employees get to fill out the remaining questions revolving around the role. If the employee has no insight into the role they immediately move forward to the next role.

# 4. Collected Information

The following chapter outlines the information we gathered on the company and the specific repair process at Computacenter Switzerland, as described in section 3.4 above.

## 4.1. Organisational Culture

Commented [TG2]: Maybe move to data collection

Overall, Computacenter strives for a high customer satisfaction. This is both true for the Computacenter Group as well as Computacenter Switzerland. It is a high priority to fulfil all the customer's needs, where possible. In Switzerland however there is less of an emphasis on existing structures. They do not necessarily impose a barrier to not deliver a service to a customer. And if there is a barrier, a workaround is found. A human connection to the customer is important.

These values are also present in the repair centre team. The nature of their service, however, does not make it always completely clear who the customer is. And this may vary case by case. In some cases, the customer is the owner of the defective device (which might be a private person or a business), it might be the manufacturer for whom we fulfil the warranty claims or in some cases it might be the reseller, who is forwarding all the defects for repair. This the mater on how customer satisfaction can be achieved.

It became apparent everyone involved is proud of their work and wants to achieve a high quality, both on the management as well as on the employee level. On one hand this means, that there is a willingness for further improvement, but on the other hand there is a certain sentiment towards the existing solution. Furthermore, there seems to be a tendency toward big changes, often technology based, which then cannot be fulfilled due to budgetary and resource restraints.

## 4.2. BPM Maturity

While this paper will not assess the Business Process Management (BPM) maturity of Computacenter in any systematic way, we were able to gain an overall impression of the overall BPM Maturity of Computacenter Switzerland. It became apparent that there is no systematic or planned approach to managing business processes in any capacity. While there is a recently introduced quality management team, which tries to improve the quality of some business processes, these efforts are at very early stages and on an ad-hoc basis. This however also

indicates that a process thinking mindset is slowly starting to emerge in certain areas of the company. All internal projects, which include changes to business processes, are triggered by a need or a pain from the business and executed without much consideration on strategic alignment or effectiveness. There is little to no process documentation other than the occasional keyword-based checklist. Nevertheless, the major value-crating and transactional processes are mostly manged in a tool with case processes capabilities, specifically the ERP System (Abacus) and the IT Service Management System (ITSM, ServiceNow). These tools make sure, that the state of each case is correctly being tracked and are assigned to the right teams for processing. Despite that, there are still many administrative tasks, that are being done outside of these tools. This includes data transformation and reporting in excel, placing purchase and dispatching orders, communication via e-mails, scanning and printing of paper-based delivery and control slips, etc.

The team concludes that Computacenter in Switzerland is still at the initial level 1 of BPM but has started the journey to level 2.

## 4.3. Strategic Alignment

The strategy within Computacenter Switzerland can be described with the following three key focus areas:

- **Scale:** Increase the company's ability to handle a growing number of clients and larger projects without compromising service quality. (Dimension: Volume)
- **Synergy:** Create a seamless and cohesive service offering that leverages the strengths of each core business area to deliver comprehensive solutions. (Dimension: Service Portfolio)
- **Long Term:** Build and maintain strong, long-lasting relationships with clients to ensure sustained business growth and customer satisfaction. (Dimension: Time)

Within the Managed Services delivery centre (of which the repair centre team is a part of) we could identify through interviews the following core processes:

- Logistics
    - Transport
    - Inbound/Outbound
    - Stock Holding
- Technology
    - Onsite
    - Repair
    - Staging
- Project Management

The SPA matrix, as shown in Figure 2, illustrates how the processes align with the company's strategy and how relevant they are to the business. From there it is evident, that the logistics process does not align well with the Scale strategy. Another problematic process is the project management when it comes to synergy. Computacenter would profit the most by focusing on these two cross sections, with a strong focus on the logistics process. Computacenter can also be satisfied that the technology process is well aligned with the company's strategy.

| | Logistics | Technology | Project Management |
|---|---|---|---|
| Scale | | | |
| Long Term | | | |
| Synergy | | | |

**Figure 2 - Strategic Process Alignment (SPA) Matrix for Computacenter Managed Service**

## 4.4. Process Overview

The following section will outline the repair process from when a defective device arrives to the point when it is being sent back to the customer.

- When the device is received from partner, the case is recorded by Log inbound team in Abacus ERP and in the Manufacturers Case System. During this, the devices is filmed from every side for evidence, in case of any damage claims.
- Check whether the device was produced by a vendor, who does repairs themselves. If so, send it to the vendor to be repaired by them. If not send the device to the Technicians team.

**Device in Warranty (IW) and no need of spare parts**

- Once it is confirmed that the device is in warranty, the technicians prioritise the cases according to the agreed upon SLAs in an excel spreadsheet. They then analyse the devices condition and proceed to repair the device if there is no need for spare parts. After repairing the device, they test the functionality and condition of the repaired device to ensure the device is fully functioning. If the device is working properly, the repaired device is sent to the customer through the log outbound team
- If a defect is found in repaired device, it is sent back to the prioritization process for further action.

**Device in warranty (IW) and needs spare parts**

- Once it is confirmed that the device is in warranty, the technicians prioritise the case according to the agreed upon SLAs, they analyse the devices condition and need some spare parts to repair the device. Then the technicians team send a request to the Parts management team. Parts Management team order the required parts and track it. After receiving the parts, they send the parts to Technicians team. Then the technicians proceed to repair the device.
- After repairing the device, they test the functionality and condition of the repaired device to ensure the device is fully functioning. If the device is working properly, the repaired device is sent to the customer through the Log outbound team.
- If the defect finds in repaired device again, it resends to the prioritization process for further action. Same as above.

**Device Out of Warranty (OOW)**
After the case is logged in from the Log inbound team, technicians prioritise the cases, analyse the devices condition, and check the warranty.

- When the device is Out of warranty (OOW) or customer damaged the device (CID), a repair quote is sent by the administration team to the customer. Then the customer can decide to proceed with the repair it, send it back to them, or to recycle the device.
- If the customer accepted the repair quote, then the technicians proceed to repair the device.
- After repairing the device, the technicians test the functionality and condition of the repaired device to ensure the device is fully functioning. If the device is working properly, the repaired device is sent to the customer through the log outbound team.
- If defects are found in the repaired device, it is sent back to the prioritization process for further action
- If the customer does not accept the repair quote it either gets sent back to the customer unrepaired or gets recycled depending on the customers wishes. This task is fulfilled by the Log outbound team.

## 4.5. Suggestions

Based on the information we have collected in the interviews and questionnaire, we have gathered the following list of possible improvements.

| # | Task | Suggestion |
|---|---|---|
| S1 | Record Case | **Preregister of cases**<br>Currently name, address, serial number and description of defect have to be typed in by inbound logistics from the delivery slip.<br>By having the case be pre-registered online by the customer, this inbound logistics only needs to associate each delivery with the right pre-registered case.<br>For B2C-Customers: Offer Web-portal for case registration<br>For B2B-Customers: Web-portal or bespoke process, such as EDI-Interface with ERP-System or API with ITSM-System |
| S2 | Record Case + Send Device back | **Photographic Record**<br>The current system relays on video and takes 40-60s per device. Focusing is very time-consuming. Replacing it with high resolution photo cameras and good lighting, which can take the picture near instantly can reduce the process time and improve the quality. |
| S3 | Record Case | **Warranty Check Automation**<br>Currently the inbound logistics manually checks the warranty status of each device on the manufacturers website by copying and pasting the serial number. Using the manufacturers API, the Case system could check warranty state automatically, after the serial number and manufacturer have been entered. |
| S4 | Record Case | **Manufacturer Case System integration**<br>Currently inbound logistics have to open the warranty case both in the local ERP system and in the manufacturers case management system, |

| | | |
|---|---|---|
| | | by manually copying and pasting all customer and information. Using either an API or robotic automation, the local ERP System could fill in the case creation form of the manufacturers automatically. |
| S5 | Customer Offer | **Catalogue Platform**<br>Currently, the creation of an offer requires the administration to search for the required spare parts on multiple vendors web shops or catalogue them via e-mail to get availability and price. By using an industry standard catalogue platform, this information would be available based on the Product EAN number requested by the technician. (e.g.: Open ICE CAT) |
| S6 | Customer Offer + Order Spare Parts | **EDI for purchase request + purchase order + invoice**<br>Currently, Computacenter requests order all spare-parts from Automatically exchange information between Abacus ERP and the supplier's system for purchasing items. So that quotes, orders and invoices automatically can be triggered. |
| S7 | Send Reminder | **Chase Automation**<br>Currently the Administration is using a report and then filters for unanswered quotes that are older than 14 days. They then manually change the status in the ERP system, which then automatically sends a reminder e-mail. By also automatically proceeding the case status, no manual intervention would be required. After 28 days, the status can then be automatically proceeded again to cancelled and the send back by the logistics is automatically triggered. |
| S8 | Order Spare Parts | **Purchase Process for Spare Parts in Abacus**<br>The Abacus ERP system includes a module to track orders, deliveries, stock, payment and customer invoicing. By Appling this module to the spare part sub-processes, the manual administrative effort can be reduced, and less human error may occur. |
| S9 | All | **Kaizen**<br>Currently there is a tendency towards big tool driven changes to drive improvements, for which there are usually not enough resources available. By introducing processes for submitting and following through with smaller changes a more continuous improvement of the process can be achieved. This includes training of the employees to identify waste, introduction of processes to submit, evaluate and execute on suggestions and it requires a budget dedicated to these kinds of small projects. |
| S10 | All | **Cross-training for multiple roles**<br>Currently, employees are trained in one or two specific roles. By cross-training and rotating them between roles, they could help out during peak demand. |
| S11 | Prioritisation | **Prioritisation in automated BI Dashboard**<br>Currently, the prioritisation of the cases is based on an excel report, that is manually evaluated. Using a scheduled report and a Business Intelligence dashboard, this could be automated and provide the relevant information on a dashboard. (e.g.: Microsoft Power BI) |
| S12 | Log Out | **Standardised Packaging** |

**Commented [TG3]:** HP: CSDP https://csdp2.hp.com/csm_new / Lenovo: NGSP https://ngsp-emea.lenovo.com/user/login

| | | |
|---|---|---|
| | | Currently, the logistics outbound team re-use the packaging, in which the devices arrived in. To reduce packaging speed and reduce damaged devices, a standardised packaging may be introduced. |

**Table 1 – Suggested solutions for current issues**

# 5. Teaching Questions

In the following section, we will provide answers to a total of 12 teaching questions from across all BPM perspectives, based on the information we have gathered.

## 5.1. Strategic Alignment

**Q1:** Create a process profile for the Repair Process and define which KPIs could be set for the said process.

<table>
<tr><td colspan="2">Name of Process: Repair Process</td></tr>
<tr><td colspan="2">Vision: To create a cost-efficient, scalable, and customer-centric device repair workflow by minimising administrative efforts through digitalisation and automation, ensuring quick turnaround times and high customer satisfaction.</td></tr>
<tr><td colspan="2">Process Owner: Gil Marini (Head of Quality Management and Operational Excellence)</td></tr>
<tr><td colspan="2">Outcome: Repaired, fully functional devices returned to customers with high quality, or unrepaired devices sent back or recycled as per customer requests.</td></tr>
<tr><td colspan="2">Trigger: A device is received at the Repair Center from a customer or partner.</td></tr>
<tr><td>Customer of the process:<br>Internal: Computacenter departments (e.g., Inbound Logistics, Technicians, Parts Management)<br>External: End customers (B2C and B2B)</td><td>Expectations of customer:<br>Efficient and timely repair of devices.<br>High-quality repair ensuring device functionality.<br>Transparent communication regarding repair status and costs.<br>Minimal delays caused by spare parts procurement or administrative inefficiencies.</td></tr>
<tr><td colspan="2">First activity: Record the device into the system in the Log Inbound team and check its warranty status.<br>…<br>Last activity: Send the repaired, unrepaired, or recycled device back to the customer via the Log Outbound team.</td></tr>
<tr><td colspan="2">Interfaces inbound:<br>• Devices from customers or partners.<br>• Warranty details from manufacturers.<br>• Spare parts from vendors.<br>Interfaces outbound:<br>• Repaired or unrepaired devices to customers.<br>• Recycling requests to appropriate facilities.</td></tr>
</table>

| |
|---|
| Required resources:<br>• ERP system for case logging and tracking.<br>• Tools and equipment for device repair.<br>• Skilled personnel in Log Inbound, Technicians, Parts Management, Administration, and Log Outbound teams.<br>• APIs or robotic process automation (RPA) for integrating warranty checks and vendor systems.<br>• Communication tools for sending quotes and updates to customers. |
| Process Performance Measures:<br>• *Turnaround Time*: Time taken from receiving the device to returning it to the customer.<br>• *Repair Success Rate:* Percentage of devices successfully repaired without requiring additional cycles.<br>• *Customer Satisfaction Score (CSAT)*: Feedback on repair quality and timeliness.<br>• *Automation Rate:* Percentage of administrative tasks automated (e.g., warranty checks, case logging).<br>• *Cost per Repair:* Total Labor and administrative cost per repaired device.<br>• *Spare Parts Lead Time*: Time taken to procure and supply required parts.<br>• *Quote Acceptance Rate*: Percentage of out-of-warranty repair quotes approved by customers. |

**Table 2 process profile for the repair process**

To effectively measure and enhance the performance of the repair centre process, a comprehensive set of KPIs is recommended, combining both existing and additional metrics. Operational Efficiency KPIs include metrics like Turnaround Time (TAT), Repair Throughput, Parts Procurement Lead Time, Workforce Utilization Rate, and Repair Workflow Efficiency to ensure streamlined operations. Quality Assurance KPIs such as First-Time Fix Rate (FTFR), Post-Repair Defect Rate, Testing Accuracy Rate, and Rework Rate focus on maintaining high repair standards. Customer Satisfaction KPIs like Customer Satisfaction Score (CSAT), Repair Quote Acceptance Rate, Status Update Timeliness, and Complaint Resolution Time aim to enhance customer experience. Cost Efficiency KPIs measure Cost per Repair Case, Administrative Automation Rate, and Spare Parts Utilization Efficiency to optimize resources. Process Reliability KPIs, including Spare Parts Availability, Warranty Verification Accuracy, and Case Logging Accuracy, ensure consistency and dependability. Environmental KPIs, such as Device Recycling Rate, Paperwork Reduction Rate, and Energy Usage, align with sustainability goals. Employee Productivity KPIs like Cross-Training Rate, Employee Satisfaction, and Training Completion Rate focus on workforce adaptability and engagement. Lastly, Strategic KPIs such as SLA Compliance Rate, Revenue per Repair Case, and Growth in Repair Volume support long-term scalability and profitability. Together, these KPIs provide a holistic framework for continuous improvement and success.

**Q2:** Provide KPIs to measure the health in the SPA matrix shown in Figure 2 in section 4.3.

**KPIs for Process Health in Strategic Alignment Matrix**

| KPIs | Logistics | Technology | Project Management |
|---|---|---|---|
| Scale | Order Fulfilment Rate, Cycle Time per Order, Warehouse Utilization Rate | System Uptime Percentage, System Scalability Index, Infrastructure Utilization Rate | Project Completion Rate, Resource Allocation Efficiency, Milestone Achievement Rate |
| Long Term | Supplier Reliability Index, Logistics Cost Per Unit, Inventory Turnover Ratio | Technology ROI, System Reliability Over Time, Innovation Adoption Rate | Project Success Rate Over Time, Knowledge Retention Index, Continuous Improvement Adoption Rate |
| Synergy | Cross-Department Coordination Efficiency, Inventory Data Accuracy, Logistics Workflow Integration Score | System Integration Score, Interdepartmental Data Accessibility, Collaborative Tool Adoption Rate | Cross-Functional Collaboration Index, Interdepartmental Communication Effectiveness, Shared Resource Utilization Rate |

**Table 3 KPIs for process health**

## 5.2. Governance

**Q3:** Which organisational structure does the Repair Center Department currently have?

The Repair Center Department at Computacenter appears to follow a functional organizational structure, where responsibilities are divided into specialised teams based on their roles within the repair workflow. The key functional teams include:

1. **Log Inbound Team** – Responsible for receiving devices, logging cases, and warranty verification.
2. **Technicians Team** – Conducts diagnostics, repairs, and post-repair testing.
3. **Parts Management Team** – Handles spare parts procurement, inventory management, and tracking.
4. **Administration Team** – Communicates with customers, generates repair quotes, and manages billing.
5. **Log Outbound Team** – Manages packaging and shipping repaired or recycled devices.

This structure allows specialisation within each team and ensures that tasks are handled by those with expertise in their respective areas. However, it can result in inefficiencies, such as siloed communication, slower decision-making, and potential bottlenecks between functional areas.

**Q4:** Which structure do you think would be most appropriate? Explain your choice.

The current structure is fine for the repair centre department. Changing the organisational structure would have an impact on the whole company. For example, changing to a process-orientated structure would impact the other departments, as people from HR and finance would have to be pulled over. Other organisational changes can also be expensive to do, as new people would have to be hired to monitor steps of the process. This is not viable for the repair centre department due to its small size.

Changing the organisational structure could help to improve efficiency, it requires though way more resources, time and effort to implement and would not be guaranteed to have a big impact on the efficiency.

## 5.3. Methods

**Q5:** Would a Kanban System work for Repair Center Process? Justify your answer.

A Kanban System is possible to implement. Within the to-do column of the Kanban a prioritisation needs to be made, to be able to adhere to the different SLAs. Customers with a promised shorter process time would need to be then prioritised higher when the device arrives and when the spare parts arrive. The following steps within the process would then use the pull system for a new device to work on. This could potentially reduce a little time when repairing or analysing the device.

**Q6:** Place the suggestions S1-S12 in a solution prioritisation matrix and argue where to start.

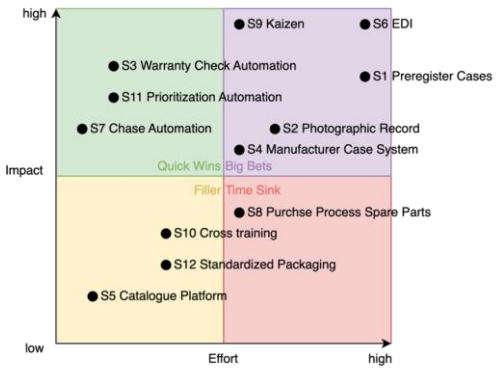


**Figure 3 – Solution prioritisation matrix for our 12 suggestions**

We would recommend starting with the suggestions in the quick win quadrant and focus on a single big bet suggestion (see Figure 3). All three quick wins are easily and quickly applied and will have a big benefit for the department. They are all focused on automation of some time-consuming administrative tasks. Out of those three its best to start with the one providing the highest impact, i.e. S3.

For the big bets its best to focus on a single one. We would argue to try S9 first. For one implementing Kaizen can be very time consuming to have all employees get used to the method. Also, it's the Big Bet that will have the highest impact with the least amount of effort. So, getting that project going should be set at a high priority. Otherwise, S6 and S1 both require collaboration with partners which is not easy. Probably its best to focus first on the solution, which is best accepted by the partners.

**Q7:** What are the four principles of Kaizen and how would you implement it for the Repair-Process.

Easier, Better, Faster, Cheaper.

First all suggestions and improvements that they currently want to implement need to become the new standard as standards are the basis for improvements. A culture of self-improvement needs to be spread throughout all employees. This generally starts from the top and trickles down. Without the top management continuously striving for improvement, the employees will not follow suit. To learn the process and teach the employees about Kaizen so that they can start using the Plan-Do-Check-Act (PDCA) cycles. This will raise awareness and train the employees on how to go about self-improvement. The whole process of implementing Kaizen may take years until its part of the companies or the departments culture.

**Q8:** Model the Repair Process according to BPMN standards.

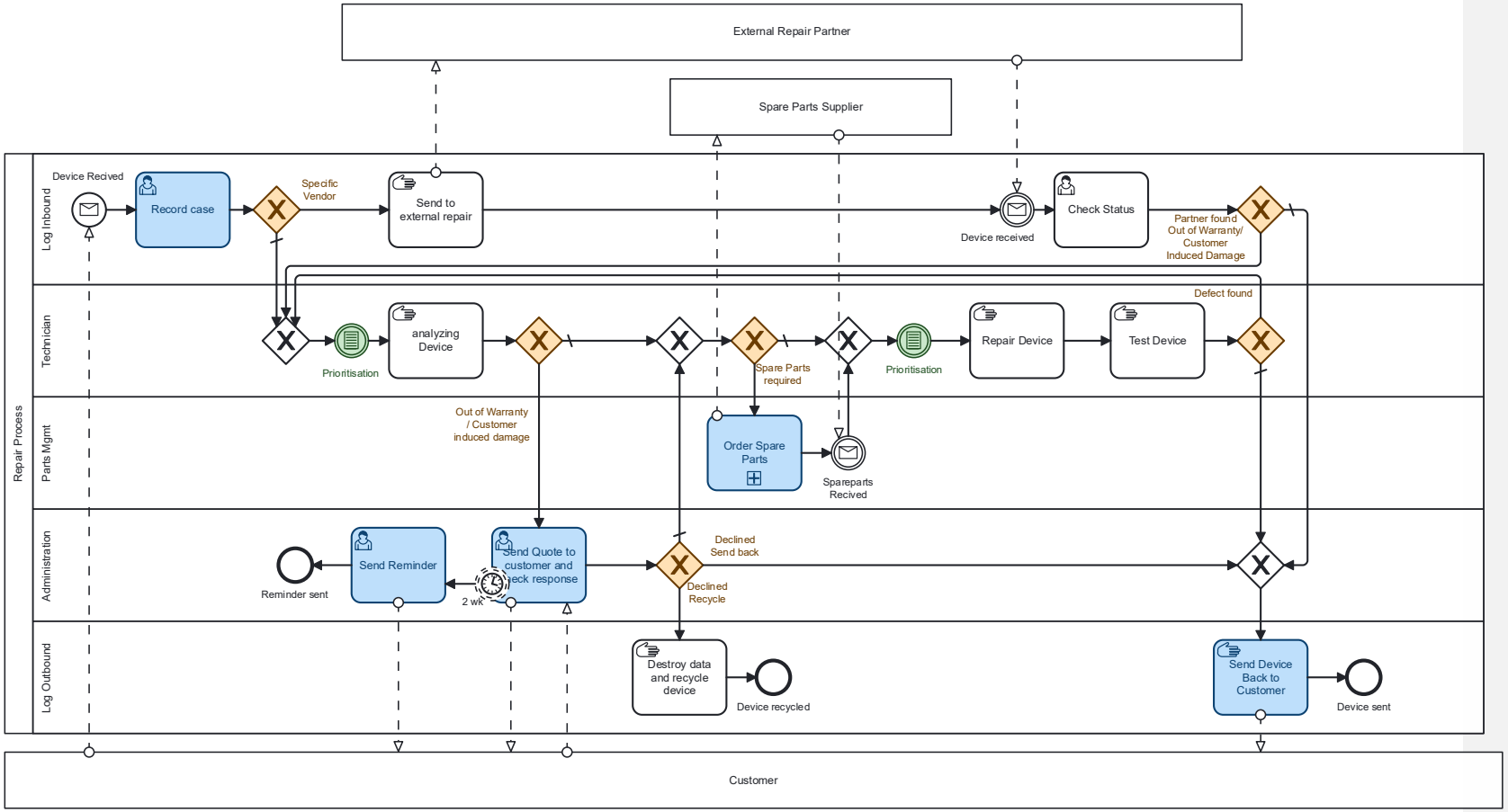


**Figure 4 - Repair Process BPMN Model**

## 5.4. Information Technology

**Q9:** What technology could be used to automate administrative tasks in the described process?

Some of our suggestions to improve administrative process steps for the Computacenter repair centre rely on exporting and transforming information from internal systems. These include the KPI calculation and the case prioritisation. Both tasks are currently based on exports out of abacus, which then are manually transformed in excel. For the automation of such tasks a *business intelligence tool like PowerBI or Tableau* could be configured to automatically load and transform the Abacus exports.

Other administrative tasks require the exchange information with third parties. Examples for these are the warranty check, the case registration in the vendors system or creation of a quote for out-of-warranty repairs. To automate those tasks, some form of integration into the third-party system is required. There are different technologies available, which could be used to achieve this. Robotic Automation could be used to automatically fill the forms of the graphical interfaces of Abacus and the third-party tools by automatically performing the mouse clicks and keyboard strokes on the workstation, which are currently entered by an employee. With Browser emulation, either the workstation or the Abacus server can fetch the information from the third parties web interface by sending the same HTTP requests as when a person would be interacting with the web interface. Where possible a full API-integration or EDI-integration can be implemented. This would require the third party to provide an interface to their system, so that Abacus can directly exchange data. Robotic automation and Browser emulation (especially if built on the workstation) require less effort for the implementation but are also more volatile to changes by the third-party, as illustrated in Figure 5 below.

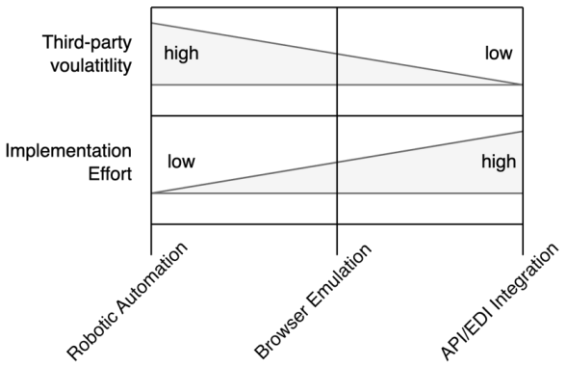


**Figure 5 - Integration Technologies**

5.5 People

**Q10:** Should Computacenter at its current state establish a BPM CoE? Explain your answer.

A full-fledged BPM Center of Excellence (CoE) may not be feasible for Computacenter at present situation. due to its low BPM maturity, and potentially limited resources.

A BPM maturity level of four is recommended to establish a BPM CoE and it requires significant resources, to design, implement, and maintain standardised processes. Computacenter operates multiple departments and relies on more manual tasks. Its current repair process struggles to balance the operational complexity with cost efficiency. Addressing bottlenecks in the repair process can often be achieved by using automation tools and improving the repair process rather than establishing a full-fledged CoE.

**Q11:** If Computacenter successfully increases its BPM maturity to the next level, what services could the BPM CoE provide to the company?

Establishing a BPM CoE in Computacenter, will centralise process governance, drive automation, reduce operational costs, and enhance collaboration, enabling to improve efficiency and prepare for future growth.

Possible services to be offered by the BPM CoE:

- Establish clear ownership of processes and responsibilities.
- BPM CoE would appoint process SMEs in each department.
- Implement the automation software to streamline the manual tasks like recording case, sending reminders and reporting and use the process monitoring tools.
- Define KPIs such as Cycle time, repairing cost per device, spare parts delivery time and customer satisfaction.
- Develop the standardized workflows and provide training on BPM tools and process improvements.
- Conduct a periodic review and identify further improvements in the process.

## 5.5. Culture

Commented [TG4]: @Timo Grünenfelder (s

**Q12:** Alyse Computacenter's culture based on the Competing Value Framework (CVF).

On the internal/external focus spectrum, the answer is quite clear. Computacenter lays a lot of value on the satisfaction of the customer and prioritises it over their internal structure and processes. Therefore, it is clearly positioned on an external focus. For the flexibility/stability spectrum, we would argue that Computacenter values flexibility more than stability. This, however, is a bit confused by the fact that the economic situation limits their capability to execute this to a full extent.

According to the CVF Matrix, depicted in Figure 6, results in an adhocracy culture for Computacenter, which can be leveraged to drive change and innovation, if done right.

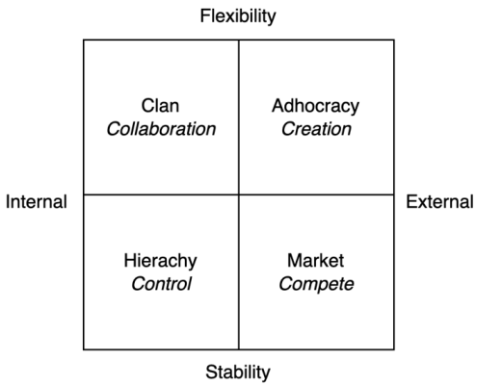


**Figure 6 – Competing Values Framework (CVF)**

# 6. Appendix A – Interview Itinerary

Interview 1
Purpose: Understanding the structure of the repair processes
Interviewee: Gil Marini – Head of Quality Management and Operational Excellence
Discussion of: Organisational Structure, Culture, Strategy, Process Tasks, Decisions, Flow Structure, Task to Role assignment

Interview 2
Purpose: Understanding the challenge
Interviewee: Mike Forrer – Head of Repair Center
Discussion of: Pain-points, Details of administrative tasks, Role descriptions, prioritisation method, SLAs, KPIs, approach (quantitative or qualitative)

Interview 3
Purpose: Evaluating the suggestions
Interviewee: Gil Marini – Head of Quality Management and Operational Excellence
Discussion of: Possible improvement actions and their feasibility and possible impact.

# 7. Appendix B – Employee Questionnaire

**Purpose of the survey:** The purpose of this survey is to collect feedback from employees who are responsible for receiving and processing customer devices for repair. We focus on the administrative work in the process (typing, scanning, paper, etc.). The answers help us to identify weak points in the process and to make suggestions for optimizing the workflow.

**Instructions:** The survey is divided into the following roles.

- Log inbound
- Spare parts management
- Administration
- Repair technician
- Log outbound

Please indicate for each role whether you work in this role. Optionally, you can also provide suggestions for the roles in which you do not work directly. Please answer the questions based on your experience in the process. The answers will be treated anonymously and confidentially.

For each of the roles, the following questions were asked:

Do you work in this role?

a) It's my primary role
b) I occasionally work on this role
c) I do not work on this role

If c) is selected, the additional following question is displayed

Do you have suggestions to improve this role?

a) Yes
b) No

If the first question is answered a) or b) or if the second answer is answered yes, the following questions are answered. The answer to each was long free text.

Describe the work you do in this role?

What are Problems and Pain points you experience with your work in this role?

What would you suggest improving your work in this role?

If the second question is answered with no, the survey skips to the next role. At the end, we provided the option to submit any other feedback.

Is there anything else you would like to tell us?

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